Seat No.: _____ Enrolment No._____

NATIONAL FORENSIC SCIENCES UNIVERSITY

M.Sc. -CS - Semester -I - JULY-2023

Subject Code: CTMSCS SII L1 **Date: 05/07/2023**

Subject Name: Network Security Lab

Time: 30 Minutes **Total Marks: 50**

Instructions:

1. **Attempt all questions.**
2. **Make suitable assumptions wherever necessary.**
3. **Figures to the right indicate full marks.**
4. **Parts of question should be attempted at the same place.**

SET-1: Roll Number ending with 1, 5, 9, 13, 17
SET-2: Roll Number ending with 2, 6, 10, 14, 18
SET-3: Roll Number ending with 3, 7, 11, 15, 19
SET-4: Roll Number ending with 4, 8, 12, 16, 20

			Marks
SET-1	Q.1	Write a Python program that implements the Caesar cipher encryption algorithm. Demonstrate its usage by encrypting and decrypting a given message.	25
	Q.2	You suspect that someone on your network is sending sensitive information over an unencrypted connection. How can you use Wireshark to capture and analyze network traffic to verify your suspicions?	
SET-2	Q.1	Set up a basic network topology in Cisco Packet Tracer consisting of two routers and two computers. Configure the routers to enable communication between the computers.	
	Q.2	You suspect that a particular device on your network is infected with malware. How can Wireshark help you identify any suspicious network activity associated with that device?	
SET-3	Q.1	A new employee brings their own device (BYOD) and wants to connect to the company network. Explain how DHCP assists in assigning an IP address to the device.	
	Q.2	A suspicious file has been downloaded on your network, and you want to analyze its behavior. How can Wireshark help you analyze the network traffic generated by the file?	
SET-4	Q.1	Implement a Python program that generates a digital signature for a file using a public-key cryptography algorithm like RSA.	
	Q.2	A suspicious file has been downloaded on your network, and you want to analyze its behavior. How can Wireshark help you analyze the network traffic generated by the file?	

END OF PAPER

Seat No.: _____ Enrolment No. 2034

NATIONAL FORENSIC SCIENCES UNIVERSITY

M.Sc. -CS - Semester -I - Jan-2023

Subject Code: CTMSCS SI L1 **Date: 18/01/2023**

Subject Name: Essentials of Cyber Security and Cyber Warfare Lab

Time: 30 Minutes **Total Marks: 50**

Instructions:

1. **Write down each question on separate page.**
2. **Attempt all questions.**
3. **Make suitable assumptions wherever necessary.**
4. **Figures to the right indicate full marks.**
5. **Parts of question should be attempted at the same place.**

34/4 = 20

SET-1: Roll Number ending with 1, 5, 9, 13, 17
SET-2: Roll Number ending with 2, 6, 10, 14, 18
SET-3: Roll Number ending with 3, 7, 11, 15, 19
SET-4: Roll Number ending with 4, 8, 12, 16, 20

			Marks
Q.1	SET-1	Using a packet sniffer to capture and analyze network traffic to reveal sensitive information.	25
	SET-2	Create the user **NFSU-your Name.** What is Windows Integrity Control (WIC)? Also set various various levels of WIC with respect to trustworthiness to your user name.	
	SET-3	Identifying vulnerabilities of your Machines.	
	SET-4	Locating target machine, and identifying its OS, open ports, types of services it runs.	
Q.2	SET-1	You receive an email from your bank telling you there is a problem with your account. The email provides instructions and a link so you can log into your account and fix the problem. What should you do?	25
	SET-2	With the help of example explain *ipconfig* and *ifconfig*?	
	SET-3	Explain CHMOD, CHROOT and CHOWN in terms of Linux OS.	
	SET-4	What is the significance of ***iptable*** in the Linux hardening? Also provide the example.	

END OF PAPER

National Forensic Sciences University, Goa Campus
Mid-Semester Examination

26 ½

Program Name – M.Sc. Cyber Security/ M.Sc. DFIS	**Sem – I**	**Date- 18-11-2022**
Subject Name –: Introduction to Forensic Science and Law		
Subject Code- CTMSCS SI P5/ CTMSDFIS SI P5		**Max. Marks- 50**
Time- 10:30 am to 12:00 pm (1.5 Hours)		
Instructions - 1) Answer all questions. 2) Assume suitable data.		

Q.1	**Answer any four of the following:**	**20 marks**
	a. What are finger prints? Enlist the various functions of Central Finger Print Bureau.	5 marks 2
	b. Discuss the various duties of forensic scientists.	5 marks 5
	c. Describe the organizational set up of Forensic Science Laboratories in India.	5 marks
	d. Write notes on the organization and functions of FBI.	5 marks 1
	e. What is INTERPOL? What are the functions of INTERPOL?	5 marks 4
Q.2	**Answer all the following:**	**15 marks**
	a. What is CCTNS and how is it related to NCRB? List out the objectives of CCTNS?	5 marks
	b. Give an account of history of development of Forensic Science in India.	5 marks
	c. Write notes on Bureau of Police Research & Development.	5 marks 3
Q. 3	**Attempt a and b:**	**15 marks**
Q.3 a.	I. State the six principles of Forensic Science and explain the same with one example each.	8 marks 6½
	OR	
	II. Write notes on any three techniques used in forensic science.	8 marks
Q3 b.	I. Explain the significance of mobile forensic laboratory in crime scene investigation.	7 marks 5
	OR	
	II. What is Forensic Science? Explain the significance of any three branches of forensic science in investigation of crime.	7 marks

End of Paper

National Forensics Sciences University, Goa Campus
Mid- semester Examination

Branch – M.Sc. Cyber Security	**Sem – I**	**Date- 17-11-2022**
Subject Name- Artificial Intelligence	**Subject Code- CTMSCS SI P4**	
Time- 1.5 Hours	**Max.Marks- 50**	
Instructions - 1) Answer all questions. 2) Assume suitable data.		

Q.1	**Solve any four**	20 marks
	a. Define **orthogonal vectors**, **orthonormal vector** and **orthogonal matrix**. Given two vectors, how can you determine orthogonality?	5 marks
	b. Consider the following set of points: **{(10, 20), (20, 40), (30, 60)}**. Find the least square regression line for the given data points.	5 marks
	c. Define instance based learning. Identify and briefly explain any machine learning model that falls under the category of instance based learning.	5 marks
	d. How can Pearson's correlation be applied to determine the relationship between two variables? Compare Spearman's and Pearson's correlation measure.	5 marks
	e. List any **two measures** used to determine the spread of data. Compute these two measures on the following data. 10, 20, 30, 40, 50	5 marks
Q.2	**Attempt all**	15 marks
	a. What is overfitting in machine learning? How can you determine if a model is overfitting? Describe any three ways to avoid overfitting.	5 marks
	b. Discuss in detail any **one** machine learning model used for **prediction**.	5 marks
	c. Apply **Naïve Bayes classification** to the data of fruits provided below to **determine the class** for an unknown data input (**Appearance** = Undamaged and **Quality** = Unripe and **Size** = Small) **Table 1 Fruit Data** **Appearance** | **Quality** | **Size** | **C** Undamaged | Unripe | Big | I Undamaged | Ripe | Small | I Damaged | Unripe | Big | II Undamaged | Unripe | Small | II	5 marks
Q. 3	**Attempt Q3a and Q3b**	15 marks
Q.3 a	**Attempt any one**	
Q.3 a	Explain any one unsupervised learning model in detail. List and explain any 5 applications of unsupervised learning model.	8 marks
	OR	

Q.3 a	Explain the different types of the machine learning system	8 marks
Q.3 b	**Attempt any one**	
Q3 b	Illustrate with a help of an example the working of a Perceptron model. List a shortcoming of the perceptron model. How can the mentioned shortcoming be overcome?	7 marks
	OR	
Q3 b	Compute the values for **True Positive**, **True Negative, Sensitivity**, **Specificity**, **Precision**, **Recall** and **F1-Score** for the following confusion matrix.	7 marks

		Predicted	
		P Sunny	N Rainy
Actual	P Sunny	TP 1984	FN 447
	N Rainy	FP 336	TN 107

Accuracy TP + TN / P + N

Precision = TP / TP + FP

Recall FP / FP + TN

Sensitivity TP/P

Specificity TN/N

F1 (2 * Precision * Recall) / Precision + Recall

National Forensics Sciences University, Goa Campus
Mid- semester Examination

Branch – M.SC Cyber security Sem – I Date- 16-11-2022

Subject Name- Web Application Security Subject Code- CTMSCS SI – P3

Time- 1.5 Hours Max.Marks- 50

Instructions - 1) Answer all questions. 2) Assume suitable data.

Q.1 Solve any four **20 marks**

a. Explain about OWAPS Zed attack proxy. 5 marks

b. Compare Authentication vs. Authorization. 5 marks

c. Write note about XXS. 5 marks

d. What is Burp sequencer? 5 marks

e. Explain the steps for secure source code review. 5 marks

Q.2 Attempt all **15 marks**

a. Recall life cycle of vulnerabilities Assessment. 5 marks

b. Write the steps about this website (http://esjindex.org/search.php?id=1) for SQL Injection. 5 marks

c. Describe about click jacking. 5 marks

Q. 3 Attempt a and b **15 marks**

Q.3 a Attempt any one

I. Explain OWAPS top ten attack. 8 marks

OR

II. Explain Security Misconfiguration and prevention method for it. 8 marks

Q.3 b Attempt any one

I. Write note on Fuzzing with burp intruder. 7 marks

OR

Ii. Elaborate SSRF and its prevention methods. 7 marks

National Forensics Sciences University, Goa Campus
Mid- semester Examination

Branch – M.SC Cyber Security & Digital Forensic Sem – I Date- 15/11/2022
Subject Name- Cyber Security Audit & Compliances Subject Code- CTMSCS SI – P2
Time- 1.5 Hours Max.Marks- 50

Instructions - 1) Answer all questions. 2) Assume suitable data.

Q.1 Solve any four **20 marks**

a. What are the controls and why are they important? 5 marks

b. Explain HIPPA in detail. 5 marks

c. Compare Risk Assessment vs Risk Analysis. 5 marks

d. What is implementation based security control. 5 marks

e. Explain IT Act Section 66. 5 marks

Q.2 Attempt all **15 marks**

a. How to Identifying the Minimum Acceptable Level of Risk and Appropriate Security in IT Infrastructure. 5 marks

b. Describe about CAAT and application control for CAAT. 5 marks

c. Explain IEC 27001. 5 marks

Q. 3 Attempt a and b **15 marks**

Q.3 a Attempt any one

I. Explain IT Act Section 43 in detail. 8 marks

OR

II. Explain GDPR and Importance of GDPR. 8 marks

Q.3 b Attempt any one

I. Elaborate seven domains of IT Infrastructure in details. 7 marks

OR

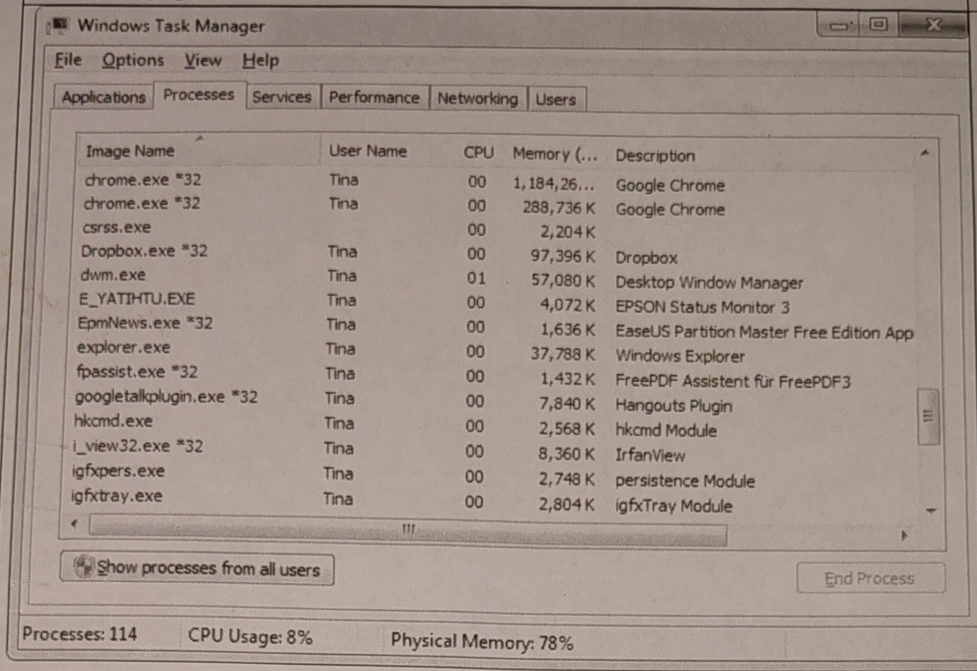
ii. List IT Audit process and types of audits. 7 marks

National Forensics Sciences University, Goa Campus
Mid- semester Examination
M.Sc. - Semester -I

Branch – Cyber Security	**Sem –** I	**Date-** 14/11/2022
Subject Name- Essentials of Cyber Security and Cyber Warfare		**Subject Code-** CTMSCS SI P1
Time- 1.5 Hours		**Max. Marks-** 50
Instructions - 1) Answer all questions. 2) Assume suitable data.		

Q.1	Attempt with reference to the Windows OS **(Q 1(a)- 1 (d)):**	**20 Marks**
	a. With reference to the Windows OS, explain the working of **Lsass.exe.**	5 Marks
	b. Explain different file permissions in Windows.	5 Marks
	c. Differentiate the Local and Domain account.	5 marks
	d. What is Windows Integrity Control (WIC)? Also compare various levels of WIC with respect to trustworthiness.	5 Marks
Q.2	Attempt with reference to the Linux OS **(Q 2(a)- 2 (c)):**	**15 Marks**
	a. Consider the following example, `gfg:x:1000:1000:main user:/home/gfg:/usr/bin/zsh` explain each field of the above format.	5 Marks
	b. What is the significance of Sticky Bit?	5 Marks
	c. Why should we not prefer **SetUID** for running a program?	5 Marks
Q. 3	Attempt any **two:**	**8 Marks**
	a. How can an attacker exploit the rootkit attack? What is the use of loadable kernel modules (LKMs) in this attack?	4 Marks
	b. With the help of suitable example, explain the 80/20 rule of functionality?	4 Marks
	c. What is the significance of **iptable** in the Linux hardening? Also provide the example.	4 Marks

Q.4	By Considering the Following Task Manager Scenario, answer the following question.	**7 Marks**



7 Marks

3/4

A. Write step by step process to identify the suspicious process.

B. Why is the chrome process considered a special case?

OR

By Considering the Following Linux Process Scenario, answer the following question. — 7 Marks

```
root         165  0.0  0.0      0     0 ?        S    20:56   0:00 [scsi_eh_2]
root         166  0.0  0.0      0     0 ?        I<   20:56   0:00 [scsi_tmf_2]
root         167  0.0  0.0      0     0 ?        I<   20:56   0:00 [kworker/0:1H
root         270  0.0  0.0      0     0 ?        I<   20:56   0:00 [raid5wq]
root         322  0.0  0.0      0     0 ?        S    20:56   0:00 [jbd2/vda1-8]
root         323  0.0  0.0      0     0 ?        I<   20:56   0:00 [ext4-rsv-con
root         389  0.0  0.6  78460  7044 ?        S<s  20:56   0:00 /lib/systemd/
root         406  0.0  0.0      0     0 ?        I<   20:56   0:00 [iscsi_eh]
root         411  0.0  0.1  97708  1880 ?        Ss   20:56   0:00 /sbin/lvmetad
root         419  0.0  0.0      0     0 ?        I<   20:56   0:00 [ib-comp-wq]
root         420  0.0  0.0      0     0 ?        I<   20:56   0:00 [ib_mcast]
root         422  0.0  0.0      0     0 ?        I<   20:56   0:00 [ib_nl_sa_wq]
root         435  0.0  0.0      0     0 ?        I<   20:56   0:00 [rdma_cm]
systemd+     532  0.0  0.3 141928  3144 ?        Ssl  20:56   0:00 /lib/systemd/
systemd+     619  0.0  0.5  71976  5980 ?        Ss   20:56   0:00 /lib/systemd/
systemd+     634  0.0  0.5  70628  5340 ?        Ss   20:56   0:00 /lib/systemd/
root         707  0.0  0.4  43092  4156 ?        Ss   20:56   0:00 /lib/systemd/
root         803  0.0  0.0      0     0 ?        I    20:56   0:00 [kworker/0:2]
```

A. Write the procedure to identify the kernel process.

B. How can we identify all existing users' processes in this scenario?